

UGA Mock 1

B.Stat./B.Math. Entrance

Instructions. Exactly one option correct per question. Time: 2 hours. Marking: +4/0/-1. No calculators.

Q1. Let $a_n = \int_0^{\pi/2} \frac{\sin((2n+1)x)}{\sin x} dx$. Then $\lim_{n \rightarrow \infty} a_n$ equals

- (A) 0
- (B) $\pi/4$
- (C) $\pi/2$
- (D) does not exist

Q2. The number of real $x \in [0, 2\pi]$ satisfying $\sin(\cos x) = \cos(\sin x)$ is

- (A) 0
- (B) 1
- (C) 2
- (D) infinitely many

Q3. Let $f : [0, 1] \rightarrow [0, 1]$ be continuous and nondecreasing with $f(0) = 0$, $f(1) = 1$. The value of $\int_0^1 f(x) dx + \int_0^1 f^{-1}(y) dy$, where f^{-1} denotes a suitable inverse, equals

- (A) $\frac{1}{2}$
- (B) 1
- (C) depends on f
- (D) $\frac{3}{2}$

Q4. Let p be an odd prime. The number of ordered pairs $(x, y) \in \{0, 1, \dots, p-1\}^2$ with $x^2 + y^2 \equiv 1 \pmod{p}$ equals

- (A) $p-1$
- (B) p
- (C) $p - (-1)^{(p-1)/2}$
- (D) $p + (-1)^{(p-1)/2}$

Q5. Let $\zeta = e^{2\pi i/7}$. Then $(\zeta + \zeta^2 + \zeta^4)(\zeta^3 + \zeta^5 + \zeta^6)$ equals

- (A) -2
- (B) 2
- (C) $\frac{-1+\sqrt{-7}}{2}$
- (D) $\frac{1-\sqrt{-7}}{2}$

Q6. The number of monic polynomials $p(x)$ with integer coefficients of degree 3 such that all roots lie on the unit circle is

- (A) 4
- (B) 6
- (C) 8

(D) 10

Q7. Let $p(x)$ be a polynomial with real coefficients such that $p(x) \geq 0$ for all real x . Then $p(x) + p'(x) + p''(x) + \cdots$ (finite sum)

- (A) is ≥ 0 for all real x
- (B) may be negative somewhere
- (C) has a real root
- (D) equals $e^{-x} \cdot \text{const}$

Q8. For positive reals a, b, c with $a + b + c = 3$, the maximum of $a^2b + b^2c + c^2a + abc$ is

- (A) 3
- (B) 4
- (C) $\frac{27}{8}$
- (D) $\frac{9}{2}$

Q9. The number of 5-digit positive integers whose digits are strictly increasing from left to right and whose digit sum is divisible by 3 is

- (A) 42
- (B) 44
- (C) 45
- (D) 48

Q10. Let $f(x) = \int_x^{x^2} \frac{dt}{\ln t}$ for $x > 1$. Then $\lim_{x \rightarrow 1^+} f(x)$ equals

- (A) 0
- (B) $\ln 2$
- (C) 1
- (D) ∞

Q11. If A, B, C are angles of an acute triangle, the minimum of $\tan A \tan B \tan C$ is

- (A) 3
- (B) $3\sqrt{3}$
- (C) $\sqrt{3}$
- (D) 9

Q12. In triangle ABC , the angle bisector from A meets BC at D and the circumcircle again at E . If $AB = 6$, $AC = 4$, $BC = 5$, then $AD \cdot AE$ equals

- (A) 20
- (B) 24
- (C) 30
- (D) 36

Q13. Let A be a real $n \times n$ matrix with $A^2 = -I$. Then

- (A) no such A exists
- (B) n must be even
- (C) $\det A = 1$
- (D) both (B) and (C)

Q14. Let $a_1 = 2$, $a_{n+1} = \frac{a_n^2 + 2}{2a_n}$. Then $\lim a_n$ equals

- (A) 1
- (B) $\sqrt{2}$
- (C) 2
- (D) diverges

Q15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(x+y) + f(x-y) = 2f(x)f(y)$ for all x, y , and f not identically zero. Then

- (A) $f(x) = \cos(cx)$ or $\cosh(cx)$
- (B) $f(x) = e^{cx}$
- (C) $f(x) = 1 + cx^2$
- (D) only $f \equiv 1$

Q16. The largest positive integer n such that $n^3 + 100$ is divisible by $n + 10$ is

- (A) 890
- (B) 900
- (C) 990
- (D) 1000

Q17. Let z_1, z_2, z_3 be distinct complex numbers on the unit circle with $z_1 + z_2 + z_3 = 0$. Then $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ equals

- (A) 6
- (B) 9
- (C) $3\sqrt{3}$
- (D) depends on positions

Q18. Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice differentiable with $f(0) = f(1) = 0$ and $|f''(x)| \leq M$ on $[0, 1]$. Then $\max_{[0,1]} |f|$ is at most

- (A) M
- (B) $M/2$
- (C) $M/4$
- (D) $M/8$

Q19. Let $S = \{1, 2, \dots, 2n\}$. The number of ways to partition S into n pairs such that each pair sums to an odd number is

- (A) $n!$
- (B) $(2n)!/2^n$
- (C) $\binom{2n}{n}$
- (D) $(2n-1)!!$

Q20. The largest integer k such that 3^k divides $10^{3^{10}} - 1$ is

- (A) 10
- (B) 11
- (C) 12
- (D) 13

- Q21.** For positive reals a, b, c with $a + b + c = 1$, the maximum of $\sqrt{4a+1} + \sqrt{4b+1} + \sqrt{4c+1}$ is
 (A) $3\sqrt{7/3}$
 (B) $\sqrt{21}$
 (C) $\sqrt{15}$
 (D) $3\sqrt{3}$
- Q22.** Let C_1, C_2 be circles intersecting at P, Q . A line through P meets C_1 at A and C_2 at B . The locus of the midpoint of AB as the line rotates about P is
 (A) a line
 (B) a circle through P
 (C) an ellipse
 (D) a circle not passing through P in general
- Q23.** Let A, B be 2×2 real matrices with $AB - BA = A$. Then
 (A) A must be invertible
 (B) $\det A = 0$ and $\text{tr } A = 0$
 (C) $A = 0$ necessarily
 (D) B must equal $\frac{1}{2}I$
- Q24.** $\lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \frac{1}{n+k} - \ln 2 \right)$ equals
 (A) 0
 (B) $-\frac{1}{4}$
 (C) $\frac{1}{4}$
 (D) $\frac{1}{2}$
- Q25.** The set of $x \in \mathbb{R}$ where $f(x) = \sum_{n=1}^{\infty} \frac{\{nx\}}{n^2}$ (where $\{\cdot\}$ is fractional part) fails to be continuous is
 (A) empty
 (B) $\mathbb{Q}$
 (C) $\mathbb{Q} \setminus \mathbb{Z}$
 (D) $\mathbb{R} \setminus \mathbb{Q}$
- Q26.** Let $p(x)$ be a monic integer polynomial of degree $n \geq 1$ such that $p(0), p(1), \dots, p(n)$ are all odd. Then
 (A) p has no integer root
 (B) p has exactly one integer root
 (C) such p exists only for $n = 1$
 (D) no such p exists
- Q27.** The number of ways to write 100 as an ordered sum of positive odd integers is
 (A) F_{99}
 (B) F_{100}
 (C) 2^{49}
 (D) 2^{50}

where F_n denotes the n -th Fibonacci number ($F_1 = F_2 = 1$).

Q28. Let S be a set of $n \geq 2$ real numbers. The number of distinct values of $|a - b|$ for $a, b \in S$, $a \neq b$, is at least

- (A) 1
- (B) $\lceil \log_2 n \rceil$
- (C) $n - 1$
- (D) $\binom{n}{2}$

Q29. The number of $x \in (0, 1)$ at which $f(x) = \sum_{n=1}^{\infty} 2^{-n} |\sin(2^n \pi x)|$ is differentiable is

- (A) 0
- (B) countably many
- (C) all but countably many
- (D) all of $(0, 1)$

Q30. Let P be a point inside equilateral triangle ABC with $PA = 3$, $PB = 4$, $PC = 5$. The side length of the triangle is

- (A) $\sqrt{25 + 12\sqrt{3}}$
- (B) $\sqrt{50}$
- (C) $\sqrt{25 + 12\sqrt{2}}$
- (D) 7

End of Paper